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## SYSTEM AND METHOD FOR MONITORING BATTERY SWELLING

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### Abstract

In an embodiment a battery pack includes a plurality of battery cells stacked in a stacking direction and forming a battery cell stack, a first end plate arranged at a first end of the battery cell stack, a second end plate arranged at a second end of the battery cell stack, connection elements connecting the first and second end plates, and a sensor arranged at the battery pack, the sensor configured to measure an extension or a shortening of a distance between the first and second end plates.

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### Background/Summary

## TECHNICAL FIELD

[0001] The present invention relates generally to a system and method for monitoring battery swelling, and, in particular embodiments, to a system and method for controlling battery swelling.

## BACKGROUND

[0002] Batteries are conventionally placed or mounted in vehicles such as cars and trucks but also in aircrafts. The number of batteries used in these vehicles may have increased over recent years. Accordingly, the industry is constantly looking to improve existing types of batteries and are in search of newer types of batteries. However, some of the more powerful batteries available or under development have a tendency to swell over time due to ageing.

## SUMMARY

[0003] Various embodiments provide a battery pack that includes a plurality of battery cells stacked in a stacking direction and forming a battery cell stack, a first end plate arranged at a first end of the battery cell stack, a second end plate arranged at a second end of the battery cell stack, connection elements connecting the first and second end plates, and a sensor arranged at the battery pack, the sensor configured to measure an extension or a shortening of a distance between the first and second end plates.

[0004] Various other embodiments provide a method for measuring a battery cell swelling of a battery pack, wherein the battery pack comprises a plurality of battery cells stacked in a stacking direction and forming a battery stack, a first end plate arranged at a first end of the battery pack and a second end plate arranged at a second end of the battery pack, connection elements connecting the first and second end plates, sensors arranged at the connection elements and a controller, and wherein the method comprises measuring, by the sensors, a value representative of an extension or a shortening of a distance of the first and second end plates.

[0005] Yet other embodiments provide a battery that comprises a plurality of battery cells stacked in a stacking direction and forming a battery cell stack, a first end plate arranged at a first end of the battery cell stack, a second end plate arranged at a second end of the battery cell stack, a plurality of tension rods connecting the first and second end plates and a plurality of sensors, each one sensor assigned to one respective tension rod and configured to measure an extension or a shortening of a distance between the first and second end plates, the extension or shortening representing a pressure between the first and second end plates, and an adjustment system comprising a motor and an adjustment device configured to adjust the distance between the first and second end plates when the measured extension or shortening exceed a threshold value.

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## Description

### BRIEF DESCRIPTION OF THE DRAWINGS

[0006] For a more complete understanding of the present invention, and the advantages thereof, reference is now made to the following descriptions taken in conjunction with the accompanying drawings, in which:

[0007] FIG. 1 shows a battery pack in a perspective view according to embodiments;

[0008] FIG. 2 shows a battery pack in a side view according to embodiments;

[0009] FIG. 3A-3B show details of a battery pack with an automatic compression adjustment system;

[0010] FIGS. 4A-4C show embodiments of a strain gauge arranged at a tension rod; and

[0011] FIG. 5 shows an embodiment method for measuring a distance between a first end plate and a second end plate of a battery stack.

### DETAILED DESCRIPTION OF ILLUSTRATIVE EMBODIMENTS

[0012] Embodiments provide a method for measuring battery cell swelling and in particular battery cell swelling with respect to state of health (SoH) and/or state of power (SoP). Further

embodiments provide a method for monitoring battery cell swelling and a method for adjusting the pressure in the cells (via adjusting end plates) in order to provide constant cell surface pressures. This is advantageous because these methods provide optimal (battery) cell performance and increased battery life.

[0013] Various other embodiments provide a battery pack or battery system related to the above the above cited embodiments. Further advantages are that such a battery pack or system is able to measure swelling and cell performance. Yet other advantages are that the disclosed battery packs provide a lightweight solution compared to other comparable battery packs.

[0014] FIG. 1 shows a battery pack **10** according to an embodiment. The battery pack **10** comprises a plurality of battery cells **100** stacked in a stacking direction S forming a battery stack **105**. A first end plate **110** is located at one end of the battery stack **105** and a second end plate **120** is located at the other end of the battery stack **105**. The battery pack **10** may comprise an insulation wrapped around the battery cells **100** (not shown) or the battery cells **100** are individually insulated by an insulation. In some embodiments a casing or housing (not shown) may at least partially surround the battery pack **10**, the battery cells **100** or the battery cells **100** and the insulation so that, e.g., the battery cells are not visible from the outside. The battery cells **100** are electrically connected to each other and the battery pack **10** comprises a plus pole and a minus pole. The poles may be arranged on the first and second end plates **110**, **120** or on the housing. For example, the plus pole may be arranged on the first end plate **110** and the minus pole may be arranged on the second end plate **120**, or vice versa. The end plates **110**, **120** may be made of an honeycomb aluminum composite.

[0015] The battery cells **100** may be solid state cells or cells with an anode made from silicon or a graphene-silicon blend. The battery cells **100** may all have the same width in a stacking direction S and may be placed in equal equidistant intervals along the stacking direction S. The surfaces of the battery cells **100** may be arranged directly adjacent to and in contact with each other. Further, a surface of a respective end cell on each side of the battery cell stack **105** may be arranged directly adjacent to and in contact with an inner surface of a respective end plate **110**, **120**.

[0016] Alternatively, coolant plates (using coolant liquid) or pads such as foams or compression pads are located between neighboring battery cells **100** and/or inner surfaces of the respective end plates **110**, **120**. The foam/compression pads and/or cooling plates may have good thermal conductivity to provide good thermal management. The foam/compression pads may be insulating, e.g., in case tab cooling is applied, in order to limit thermal runaway propagation.

[0017] The battery cells **100** may be series connected, i.e., the plus node of one cell is connected to the minus node of the next cell. Alternatively, the battery cells **100** may be connected in parallel. The cells **100** may have the nodes on the top or bottom surfaces (facing in stacking direction S) or on the side surfaces (facing orthogonal to the stacking direction S). Some cell types **100** may require constant pressure (such as certain types of solid state cells), while other cell types may tolerate some pressure change.

[0018] The first and second end plates **110** and **120** (sometimes also called pressure plates) are connected to each other via connection elements or means for connecting the end plates **130**. The connection elements **130** may be connection rods, bars, struts or bolts or tension rods, bars, struts or bolts. For example, the connection elements **130** are threaded or smooth (without a thread) tension rods. The two end plates **110**, **120** may be connected by a plurality of connection elements **130** such as 2, 3, 4, 5, 6, 8 or more connection elements **130**. The connection elements can comprise aluminum, steel, or carbon fiber rods. The rods should have a tensile strength in the range of between 100 MPa to 700 MPa, 150 MPa to 620 MPa, or 250 MPa to 500 MPa.

[0019] The connection elements **130** may permanently fixed to the end plates **110**, **120**. For example, they might be integrated with, combined with or soldered to both of the end plates **110**, **120**. Alternatively, the connection elements **130** may be permanently fixed to one of the end plates **110**, **120** and may be removably fixed to the other one of the end plates **110**, **120**. Yet, in another

alternative embodiment, the connection elements **130** may be removably fixed to both end plates **110, 120**. For example, the first and second end plates **110, 120** may have holes or openings through which the connection elements **130** are guidable or guided. In an embodiment, only one of the first and second end plates **110, 120** has holes and the other does not. The connection elements **130** may have a thread or a hole at at least one end.

[0020] The connection elements **130** may extend beyond the first and/or second end plate **110, 120** on the outer surfaces opposite to the inner surfaces where the battery cells **100** are located. The connection elements **130** may be adjustably fastened, by a fastener **140**, to the first and/or second end plates **110, 120** for tensioning them. The fasteners **140** may be nuts, clips, brackets, etc.

[0021] FIG. **2** shows a side view of a battery pack **10** according to an embodiment without an automatic compression adjustment so that no dynamic adjustment of the end plates **110, 120** is possible. The battery pack **10** may be similar to the battery pack of FIG. **1**.

[0022] The first and second end plates **110, 120** hold the battery cell stack **105**. Four tension rods **130** (or bolts or bars) extend between and beyond the end plates **110, 120**. Fasteners **140** fix the end plates **110, 120** to each other via the tension rods **130**. The distance of the end plates **110, 120** may be adjustable (although not automatically) relative to each other via the fasteners **140**.

[0023] Moreover, the battery pack **10** shows sensors **150** arranged at/on/in the tension rods **130**. The battery pack **10** may comprise one sensor **150** or a plurality of sensors **150**. For example, the battery pack **10** has four tension rods **130** each having one sensor **150** assigned, e.g., each rod has a sensor **150** attached. In an alternative embodiment only some of the tension rods **130** have an assigned sensor so that the battery pack **10** comprises only one sensor **150** or two sensors **150**.

[0024] The sensor **150** is configured to measure a pressure between the end plates **110** that is a result of the tension in the tension rods **130**. A base pressure may be established by tightening the fasteners **140** after the battery pack **105** is assembled. The pressure may further be generated or increased by swelling of the battery cells **100** and in particular by swelling of the surfaces of the battery cells **100** facing each other. The battery cells **100** may permanently swell according to ageing. Battery cells **100**, particularly lithium-ion batteries and solid state batteries, can swell under certain conditions due to a process called outgassing. This happens when the battery cell **100** is overcharged, discharged too quickly, exposed to high temperatures, physically damaged or just became less efficient (over time). For example, when the battery cell **100** is overcharged, the excessive current leads to a buildup of heat, which can cause the electrolyte in the battery to break down and produce gas. The produced gas exercises pressure on (all) battery cell walls (and in particular the top and bottom surfaces of the cells **100** in stacking direction **S**). The generated pressure on the top and bottom surfaces of battery cells **100** in turn generates tension on the tension rods **130**. For example, a minimum pressure could be 200 kPa, 300 kPa or 400 kPa. The threshold pressure could be 1 MPa, 1.5 MPa, 2 MPa, or even 3 MPa.

[0025] The sensor **150** may be arranged on/in/at or between tension rod(s) **130**. Alternatively, the sensor **150** may be arranged on at least one of the battery cells. The sensor **150** may be a strain gauge. The strain gauge may measure the pressure between the end plates **110, 120** by measuring the strain of the connection elements (tension rods). The strain gauges may comprise a wire pattern, a capacitor, a ring oscillator, an inductive element, a membrane, a crystal, or a strain gauge.

[0026] The sensor(s) **150** may be connected to a control system (CS) or a battery management system (BMS) **160** comprising a processor **161** such as a microprocessor. The control system (CS) is configured for pressure control and may be independent and distinct of the BMS. For example, each battery pack can have an assigned CS **160** but the entire battery comprising several battery packs may have only one BMS **160**.

[0027] The CS/BMS **160** may be configured to calculate the compression force between the plates **110, 120**. In order to calculate the compression force three values need to be known: 1) an extension or elongation (along direction **L**) of a respective tension rod **130**, 2) Youngs Modulus of the respective tension rod **130** and 3) the cross sectional area of the respective tension rod **130** at

the location of the sensor **150**. Alternatively, 3) can be replaced by a ratio between the (top/bottom) surfaces of a cell and the cross sectional area of the tension rod **130**. The first item is measured by the sensor **150** and the two last items are known so that the sensor **150** or the CS/BMS **160** can calculate the strain, pressure and/or the compression force.

[0028] Accordingly, each sensor **150** is configured to measure the elongation or extension of the tension rod **130** (or an equivalent of that value) and to report this value to processor **161** of the CS/BMS **160** and the processor **161** is configured to calculate a pressure (value) and/or a compression force (value). In various embodiments the sensor has the capability to calculate the pressure and/or the compression force.

[0029] The processor **161** is further configured to compare the measured/calculated pressure value or compression force value to a threshold pressure value or compression force value. If the measured value exceeds/passes the threshold value (above/below) the processor **161** of CS/BMS **160** is configured to issue an indication that the measured value has passed the threshold value. The indication may be an illumination, a sound signal or other notification so that a user is able to know that some action is required.

[0030] The battery pack **10** of the embodiment of FIG. 2 may not have the capability for adjusting the end plates **110**, **120** because they are arranged in a permanently fixed distance. Accordingly, this arrangement is suitable for certain types of battery cells **110** that tolerate some degree of pressure. The sensors **150** in this arrangement may indicate when the battery pack **10** reaches its end of life and needs to be replaced with a new one.

[0031] Alternatively, the distance of the end plates **110**, **120** of the battery pack **10** in FIG. 2 may be manually adjustable. This arrangement may be suitable for battery cells **110** that do not need frequent adjustment. The sensors **150** in this arrangement may indicate when the pressure passes the threshold value and a user may loosen the fasteners **140**, e.g., at one of the end plates **110**, **120**, in order to mitigate the pressure and in order to move the pressure value away from the threshold value back in the allowed pressure region.

[0032] FIGS. 3A-3B show side views of an end portion of a battery pack **10** according to an embodiment with automatic compression adjustment so that a dynamic adjustment of the end plates **110**, **120** is possible. The battery pack **10** may be similar to the battery pack of FIGS. 1 and 2.

[0033] As in FIG. 2, each sensor **150** is configured to measure an elongation of the tension rod **130** or a pressure between the end plates **110**, **120** by an increase in battery cell **110** size due to swelling. Each sensor is configured to report the measured value to the processor **161** of the CS/BMS **160** and the processor **161** is configured to compare the measured value to a threshold value. If the measured value exceeds/passes the threshold value, the processor **161** is configured to notify an adjustment system **170** so that the first plate **110** and/or second plate **120** need(s) to be adjusted in order to relieve, reduce or alleviate the pressure between the first and second plates **110**, **120** and the adjacent surfaces of battery cells **100**.

[0034] The adjustment system **170** may comprise a motor **175** and an adjustment device or means for translating the adjustment action **178** of the motor **175**. The motor **175** may be a stepper motor or a continuous servo motor. The adjustment device **178** may be a series of gears that moves (e.g., turns), a fastener **140** (e.g., a nut) on a tension rod **130** in order to adjust or loosen (one of) the plates **110**, **120**. Each tension rod **130**/sensor **150** may have its own motor **175** and its own adjustment device **178** assigned. In an embodiment, only one motor **150** and a plurality of adjustment devices **178** are employed. In a further embodiment, only one motor and one adjustment device is employed in order to operate all fasteners. In the latter cases, the motor **175** may have a coupling or clutch configured to engage all or some (e.g., a single) adjustment device(s) **178**.

[0035] In operation, each sensor **150** measures the elongation of a respective tension bar or the pressure between the first and second plates **110**, **120**. If it detects a strain (above a certain threshold value) it communicates the detection to the processor **161** of the CS/BMS **160**. The processor **161** in turn notifies the adjustment system **170** and in particular its motor **175**. The

processor **161** indicates how much adjustment is needed. The motor **175** operates the adjustment device **178**, which in turn moves the fastener **140**. For example, the adjustment device **178** is configured to loosen the fastener **140** (e.g., nut) so that the second end plate **120** is able to move away from the first end plate **110** and thereby mitigates the pressure in battery cells **100** and on the surfaces of the battery cells **100**.

[0036] In one embodiment, the adjustment device **178** comprises two gears, a first (smaller) gear **176**, which is fixed directly to or on a shaft or spindle of the motor **175**, and a second (larger) gear **177**, which is fixed to the faster **140** (e.g., nut) so that the first gear **176** is configured to directly move or turn the second gear **177** (via, e.g., a step down ratio). The shaft or spindle of the motor **175** may be arranged parallel to a direction of the connection element **130** (e.g., tension rod).

[0037] Alternatively, the step down ratio can also be achieved with a second (additional) shaft or spindle (on which gears are arranged) between the motor shaft or spindle and the fastener **140** of the connection element **130** in order to provide the step down ratio. In this embodiment the first gear may directly connect to and move a third gear of the additional shaft or spindle and a fourth gear of the additional shaft or spindle may directly connect or move the second gear.

[0038] The battery pack **10** may further comprise a bearing **190** arranged at the connection element **130** between the fastener **140** and the second end plate **120**. The bearing **190** may be an axial bearing or a thrust washer.

[0039] FIG. 4A shows an embodiment of the sensor **150** arranged at a connection element **130** such as a threaded tension rod. The sensor **150** may be a discrete element such as a load cell, which is permanently fixed to, removably fastened to, or integrated in or on a surface of the connection element **130**. For example, the sensor **150** may surround a portion of the surface area of the connection element **130**. Alternatively, the sensor **150** may be a foil glued to a surface of the connection element **130**. In various embodiments the sensor **150** is or comprises a strain gauge **155** having a measurement pattern **153**. For example, the sensor **150** is a strain gauge **155** with embedded measurement pattern **153** (e.g., embedded in an isolation foil). The measurement pattern **153** may comprise an electrically conductive path and terminals **151**. The electrically conductive path between the terminals **151** may be a wire or filament. The sensor **150** may further comprise a power supply configured to supply power to the terminals **151**. The power supply may additionally comprise a voltmeter and/or an amperemeter. The voltmeter and/or amperemeter measures the electric potential or the electric current between the terminals **151**. The CS/BMS **160** may provide the power to the sensor **150** in order to perform the measurement(s). Alternatively, the power is provided from some other power sources and the controller **161** of the CS/BMS causes the other power source to power the sensor **150**.

[0040] The sensor **150** sends the electrical measurement(s) back to the CS/BMS **160**. The processor **161** of the CS/BMS **160** calculates a resistance value and compares the measured resistance value with a (predetermined) threshold resistance value or only compares the current value with a predetermined current value. Alternatively, the sensor **150** calculates the resistance value and sends it over to the CS/BMS **160** for the comparison. In another embodiment the sensor **150** calculates the resistance value and comparison and just sent over the final result for the CS/BMS **160** in order to take action.

[0041] For example, in operation, when connection element **130** is extended it narrows a width of the wire or filament **153** and increases a length of the wire or filament **153**. When electrical power or electric current is applied (via the terminals) a measured resistance/current (as equivalent of the extension) between the terminals **151** is increased relative to an original resistance/current between these terminals. As mentioned above, the strain can be measured via other options, e.g., a capacitor or an inductive element.

[0042] The connection element **130** (e.g., rod) may have threads at both ends of the element **130**. One thread **132** may be longer than the other thread **134**, e.g., twice as long or three times as long. Alternatively, both ends may comprise threads with the same thread lengths.

[0043] FIGS. 4B-4C show an alternative embodiment to the embodiment of FIG. 4A. FIG. 4B shows an exploded view of the embodiment and FIG. 4C an assembled one. Here, the sensor **150** is not arranged at or on a surface of the connection element **130** but connected between two (same) connection elements (threaded rods) or a connection element **130** and a threaded bolt **135** or a bolt integrated in one of the end plates, etc. The sensor **150** may comprise at least one strain gauge **155** arranged on a surface or integrated in an aluminum sensor **150**.

[0044] In various embodiments the sensor **150** is not arranged on the connection elements but on the battery cell or a plurality of battery cells.

[0045] FIG. 5 shows a method **500** for measuring a pressure between a first end plate and a second end plate of a battery stack in a battery pack, wherein the battery stack comprises a plurality of cells stacked in a stacking direction. The battery pack further comprises connection elements such as (tension) rods or bars connecting the first and second end plates. The battery pack comprises at least one sensor such as a strain gauge for measuring an elongation of the assigned connection elements (e.g., as an indication of electrical current) and/or a pressure of the end plates. A sensor may be arranged at or between connection elements. The battery pack may have several connection elements such as four connection elements and several sensors, each one sensor assigned or arranged at a respective one of the connection elements.

[0046] In a first step, at **502**, the sensor measures a change in elongation of at least one connection element and/or a pressure between the first and second end plates. A change in current, resistance, capacity, inductivity is indicative of an elongation change or a pressure change. For example, a change in electric current can be indicative of a pressure change. In various embodiment, in order to measure the change in pressure between the first and second end plates, the sensor may measure the electric current (value) and/or based on this a resistance (value) of a wire.

[0047] In a second step, at **504**, the sensor or a control system or a battery management system compares the measured or calculated pressure value with a predetermined pressure threshold value. The predetermined threshold value is based on the base (original) pressure existing between the first and second end plates after the battery pack is assembled and has passed quality check. The predetermined threshold also takes into account an acceptable margin or tolerance in which no action is required, i.e., how much the pressure can increase without causing a concern for the battery pack after assembling or during the life of the battery pack.

[0048] If the measured value exceeds the (upper or lower) threshold value the controller of the battery management system, at **506**, will issue a notification. The notification may be issued as an alarm (e.g., an illumination of a light at the battery pack), at **507**, so that a user is aware that the current battery pack must be changed with a replacement (e.g., new) battery pack or that the fasteners at the end of the connection elements at an end plate need to be manually loosened.

[0049] Alternatively, the notification may be sent as an electrical signal, at **508**, to an adjustment system arranged at the battery pack. The electrical signal may trigger or start, at **510**, a motor of the adjustment system in order to perform an adjustment action (loosen a fastener at an end plate at an end of the connection element). The motor may adjust the distance (e.g., increase) between the first and second end plates so that the pressure on the battery cells is mitigated. This may be done by the adjustment device of the adjustment system. The adjustment device may comprises gears operating (e.g., unscrewing) a nut.

[0050] After **510**, at **512**, the CS/BMS may control the pressure via the sensors again in order to verify that the proper pressure values are achieved. The motor may be a servo motor being able to adjust the pressure continuously or a stepper motor that is able to adjust the pressure in discrete steps. If the proper pressure values are not achieved, the method repeats the method **600**. In various embodiments, the CS/BMS of the battery stack controls the pressure as much as at least once per minute, at least once per hour, at least once per day, or at least once per week.

[0051] A variation of this method could also be used when the battery is tested after assembling during manufacturing. If every connection element (tension rod or bar) has a sensor, each sensor

may provide pressure information. For a properly manufactured battery pack all sensors should indicate the same pressure (and in particular should not pass the threshold value). If one or more sensors show an error (too high or too low) the distance between the end plates may be adjusted via manual (or automatic) adjustment of the fasteners so that the pressure between the end plates is moved to a desired value.

[0052] Moreover, during operation, all sensors should report about the same pressure. If only one sensor of the sensors indicates a violation of the threshold value the CS/BMS should indicate a failure or issue a warning. Similarly, if the sensors (e.g., one sensor compared to the other sensors) show a substantially large difference in pressure (e.g., 30% or 40%) without violating the threshold value, the CS/BMS should also indicate a failure or issue a warning.

[0053] While this invention has been described with reference to illustrative embodiments, this description is not intended to be construed in a limiting sense. Various modifications and combinations of the illustrative embodiments, as well as other embodiments of the invention, will be apparent to persons skilled in the art upon reference to the description. It is therefore intended that the appended claims encompass any such modifications or embodiments.

## Claims

1. A battery pack comprising: a plurality of battery cells stacked in a stacking direction and forming a battery cell stack; a first end plate arranged at a first end of the battery cell stack; a second end plate arranged at a second end of the battery cell stack; connection elements connecting the first and second end plates; and a sensor arranged at the battery pack, the sensor configured to measure an extension or a shortening of a distance between the first and second end plates.
2. The battery pack of claim 1, wherein the sensor is a strain gauge configured to measure strain of a respective connection element.
3. The battery pack of claim 1, wherein the sensor comprises a plurality of sensors, and wherein each sensor is assigned to or arranged at a respective connection element.
4. The battery pack of claim 3, wherein the connection elements are exactly four connection elements.
5. The battery pack of claim 1, wherein the connection elements are tension rods, and wherein each sensor surrounds a portion of a surface area of a respective tension rod.
6. The battery pack of claim 1, wherein the connection elements are tension rods, and wherein each sensor is arranged between two tension rods.
7. The battery pack of claim 1, wherein the connection elements are tension rods and bolts, and wherein each sensor is arranged between a tensions rod and a bolt.
8. The battery pack of claim 1, wherein the sensor is configured to: measure the extension or the shortening; calculate strain based on the extension or the shortening; compare the calculated value with a threshold value; and provide a notification when the calculated value exceeds the threshold value.
9. The battery pack of claim 1, further comprising a measuring system or a battery management system configured to: receive a strain measurement from the sensor; compare a measured value with a threshold value; and provide a notification when the measured value exceeds the threshold value.
10. The battery pack of claim 1, further comprising a measuring system or a battery management system configured to: provide power to the sensor; receive a measured current value or a measured resistance value; compare the measured value with a threshold value; and provide a notification when the measured value exceeds the threshold value.
11. The battery pack of claim 1, further comprising an adjustment system, wherein the adjustment system is configured to: receive a notification that a measured value related to pressure between the first and second end plates exceeded a threshold value; and adjust the distance between the first and

second end plates.

**12.** A method for measuring a battery cell swelling of a battery pack, wherein the battery pack comprises a plurality of battery cells stacked in a stacking direction and forming a battery stack, a first end plate arranged at a first end of the battery pack and a second end plate arranged at a second end of the battery pack, connection elements connecting the first and second end plates, sensors arranged at the connection elements and a controller, the method comprising: measuring, by the sensors, a value representative of an extension or a shortening of a distance of the first and second end plates.

**13.** The method of claim 12, wherein measuring the extension or the shortening comprises measuring a current value or a resistance value at the sensors.

**14.** The method of claim 12, further comprising: comparing the measured value with a threshold value; and issuing a notification when the measured value exceeds the threshold value.

**15.** The method of claim 14, further comprising: automatically adjusting the distance between the first and second end plates after the notification is issued.

**16.** The method of claim 14, further comprising: replacing the battery pack with a new battery pack at a vehicle when the notification is issued.

**17.** A battery comprising: a plurality of battery cells stacked in a stacking direction and forming a battery cell stack; a first end plate arranged at a first end of the battery cell stack; a second end plate arranged at a second end of the battery cell stack; a plurality of tension rods connecting the first and second end plates; and a plurality of sensors, each one sensor assigned to one respective tension rod and configured to measure an extension or a shortening of a distance of the tension rods fastened to the first and second end plates, the extension or shortening representing a pressure between the first and second end plates; and an adjustment system comprising a motor and an adjustment device configured to adjust the distance between the first and second end plates when the measured extension or shortening of the tension rods exceed a threshold value.

**18.** The battery of claim 17, wherein the tension rods are permanently fixed to the first end plate and releasably fixed to the second end plate.

**19.** The battery of claim 17, wherein the adjustment device comprises a plurality of gears between the motor and the tension rods, wherein a first gear of the plurality of gears sits on a spindle connected to the motor, wherein a second gear of the plurality of gears sits on a respective tension, wherein the first gear is configured to move the second gear, and wherein the second gear is configured to move a nut on the respective tension rod in order to move the second end plate away from the first end plate.

**20.** The battery of claim 17, further comprising a controller configured to receive the measurement and to send a notification to the adjustment system when the measured extension or shortening exceeds the threshold value.

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